

UAI101: Foundation of Machine Intelligence: Concepts, Techniques and Applications

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AI Essentials: Definition and scope of AI, History and evolution of AI, AI vs. Human Intelligence, AI in engineering and technology.

Intelligent Agents and Environment Interaction: Definition and Types of Intelligent Agents, Agents and Environments: How agents perceive and act.

Knowledge Representation and Reasoning: Introduction to Knowledge Representation, Logic-based approaches: Propositional and Predicate Logic, Rule-based Systems and Expert Systems

Introduction to Data and its analysis: Types of data: Structured, Semi-structured, Unstructured, Data Streams, Statistical Data types, Sampling.

Data Analysis Methods: Descriptive Analysis, Exploratory Analysis, Inferential Analysis, Predictive Analysis.

Machine Learning Essentials: Introduction to machine learning, Types of learning (Supervised, Unsupervised, Semi-Supervised, and Reinforcement Learning, Self - Supervised Learning) Introduction to Neural Networks and Deep Learning.

Recent Advancements in AI: Explainable AI (XAI), Edge AI, Generative AI, AI for Social Good (Applications of AI in healthcare, education, climate change, and sustainability), AI Security (Protecting AI systems from attacks and ensuring secure AI development)

Case studies of AI-driven innovations: Smart Speakers, Self-Driven cars, Smart Agriculture.

Text Books / Reference Books:

1. M. Mohri, A. Rostamizadeh, and A. Talwalkar, *Foundations of Machine Learning*, 2nd ed. MIT Press, 2018.
2. E. Alpaydin, *Introduction to Machine Learning*, 4th ed. MIT Press, 2020.

Course Learning Objectives

1. Understand and Describe Fundamental AI Concepts and Intelligent Agents
2. Apply Knowledge Representation and Data Analysis Techniques
3. Differentiate and Implement Machine Learning Techniques
4. Evaluate and Discuss Recent Advancements and Innovations in AI

SEMESTER II